Spec: Tau "The Pipe" / Haible 20-stage phaser emulation

Open-source (MIT), non-commercial plugin. Prepared 6 Nov 2026 for an implementing agent. Fidelity is defined by the measurements in Section 8.

1. Purpose and scope

Build an open-source (MIT), non-commercial plugin reproducing the Tau "The Pipe" phaser as redesigned by Jürgen Haible: a 20-stage ("20-pole") transistor-ladder all-pass phaser with 1 V/octave pitch control, resonance/feedback to self-oscillation, an internal triangle LFO, and two complementary outputs giving pseudo-stereo. Fidelity is judged against the current Random*Source hardware edition by the measurements in Section 8.

Out of scope: emulating the onboard power supply, the CV mixer add-on, panel artwork, or any Random*Source PCB artwork/layout. The repo contains no schematic, Gerber or PCB material from Haible or Random*Source.

2. Lineage and reference hardware

- **Original:** the Tau "The Pipe" was developed in 1975 by Dennis Colin and Ken McNeill and released by Tau Systems (Newton, New Hampshire) around 1980. Colin also designed the ARP 2600 and Aries synths. Though sold as a "flanger", it is a phaser. The original used potted "PD-10" transistor-ladder modules, which is why DIY recreations were impractical until Haible's redesign.
- **Haible redesign (2007):** Jürgen Haible (1961–2011) redrew the circuit with the potted ladder circuitry moved onto the main PCB, avoiding obsolete parts (e.g. a dual FET). He described it as inspired by Aries, ARP and Tau circuits and closely related to the ARP Quadra phaser he had cloned earlier. Twenty all-pass stages built from matched transistor pairs (CA3046/CA3086 arrays, or LM394/SSM2210/MAT-02 pairs, or hand-matched discretes); 1 V/oct tracking with a 560 Ω tempco (original Tau: 1.87 kΩ); a "1/x function generator" in the control path that he warned is sensitive to supply voltage (±15 V design; ±12 V gives the wrong sweep range).
- **Random*Source editions (2015/16, 2021 Euro, 2026):** Random*Source states it acquired all rights to Haible's electronic heritage. Their boards use 10 × THAT300 or 20 × BCM847DS matched pairs, an LM3046 in the 1/x generator, and add a -1 V/oct input, attenuverting CV, a HARD/SMOOTH switch (high-pass on the LFO CV), a NORMAL/COLOR switch (inverts the effect signal), and NORMAL/ALTERNATIVE outputs for pseudo-stereo. Their listings describe the Euro module as "18-pole"; Haible's own page and the 2026 edition say 20 transistors/poles. Treat N as a build-time constant (default 20) and confirm by counting notches in the measured response.
- **Related patent:** US 3,969,682 "Circuit for dynamic control of phase shift" (filed Oct 1974; same ARP-era design community) describes the current-controlled all-pass stage with pole $\omega = I_b \cdot K / C$. It is expired and is a useful theoretical reference for the control law even though the Tau ladder uses junction dynamic resistance rather than an OTA.
- **Reference instrument to buy:** Random*Source "Haible TAU Phaser" Eurorack (18 HP, ≈US$400 at Perfect Circuit/Schneidersladen) or the 2026 "TAU Phaser 'The Pipe' 2026" with two audio and two CV inputs. Needs ±12 V Euro power and a DC-coupled interface or a calibrated CV source for the measurements in Section 5.

3. Signal-flow model

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in → input attenuator → [feedback sum] → AP1 → AP2 → … → AP_N (N=20) → wet wet → (× feedback gain, soft-limited) → back to [feedback sum] NORMAL out = dry + wet (or dry − wet with COLOR/INVERT) ALTERNATIVE out = complementary mix (dry − wet, or a different tap; determine by measurement) control: PITCH + 1V/oct CV + attenuverted CV + (LFO × DEPTH, optionally HPF'd by SMOOTH) → exponential converter → 1/x generator → ladder control current → all stages share one pole frequency f_c(t)
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Blocks to model explicitly

- **All-pass ladder.** N first-order all-pass sections, nominally identical pole frequency f_c. The hardware's character comes from (a) the sheer stage count — up to N/2 = 10 notches in the summed response, giving the "through-zero flanger" impression users report, (b) the per-stage mismatch from transistor matching (THAT300/BCM847 are tightly matched; CA3046 builds less so), and (c) the junction-resistance variable element, whose small-signal resistance r_e = V_T / I makes each stage mildly level-dependent (the all-pass pole moves with signal amplitude, producing distortion and "squash" at high input). Implement as TPT first-order all-pass sections; add a per-stage pole offset vector (measured) and an amplitude-dependent pole-shift term fitted to the THD sweep.
- **Control law.** 1 V/oct nominal (tempco-compensated), followed by the 1/x generator. Do not assume pure exponential: measure f_c vs. CV at 0.5 V steps over the full range and fit (expect exponential over most of the range with compression at the extremes). Haible notes tracking is only approximate and is not critical because the device is a filter, not an oscillator — but in self-oscillation it does track, so the fit must be good enough for the oscillator to stay within ±25 cents over 4 octaves if the hardware does.
- **Feedback/resonance.** Global feedback around the whole chain, up to and including self-oscillation. The "nice/screaming" trimmer option in Haible's build notes and the factory resonance trim indicate a soft limiter in the loop; characterise the oscillation amplitude and waveform and fit a tanh/soft-clip stage. Sign of the feedback (positive vs. negative) determines whether peaks or notches resonate; determine from measurement.
- **Outputs.** NORMAL and ALTERNATIVE are complementary dry/wet sums (one yields notches where the other yields peaks), which is what produces the pseudo-stereo swirl. COLOR/INVERT swaps the sign on the NORMAL path. Measure all four combinations.
- **LFO.** Triangle, fairly fast, with UP/DOWN LEDs indicating direction. SMOOTH = high-pass filtering of the LFO CV (subtle). Rate range and the optional slow mode (Random*Source: L* jumper) must be measured. Depth control blends between 100 % LFO and 100 % PITCH+CV (it is a crossfade, not a simple depth multiplier — model it that way).
- **Phase/Vibrato mode** (Haible and PedalPCB builds): a switch that removes the dry path so the chain acts as a pitch-modulating all-pass (vibrato). Include as a mode.

4. Parameter table (plugin)

Parameter	Range / type	Hardware equivalent
Pitch (centre frequency)	Displayed in Hz (f_c of the all-pass stages), range from measurement (expect roughly 20 Hz–20 kHz across the CV span)	PITCH knob
Resonance	0–100 % of loop gain, continuing into self-oscillation above the measured threshold	FEEDBACK / RESONANCE
LFO Rate	Hz, log; range from measurement; optional Slow mode	RATE, L* slow mode
LFO Depth	0–100 % crossfade LFO ↔ Pitch+CV	DEPTH
LFO Smooth	Off / On (HPF on LFO CV)	HARD/SMOOTH
Color (invert wet)	Normal / Color	NORMAL/COLOR, INVERT toggle
Output mode	Normal / Alternative / Stereo (Normal L, Alternative R)	NORMAL & ALTERNATIVE jacks
Mode	Phase / Vibrato	Phase/Vibe switch
Stages N	Build constant, default 20 (12/16/18/20 selectable for experiments)	PD-10 ladder count

Parameter	Range / type	Hardware equivalent
CV in	Sidechain / host modulation, 1 V/oct scaled, plus attenuverted second input	1V/Oct, CV ATT
Input level	-∞ … +6 dB	INPUT attenuator
Drive/matching	Advanced: per-stage mismatch amount (0 = ideal, 1 = measured)	transistor-array choice

5. Reference instruments and data to capture

- Random*Source Tau Phaser (Euro) powered from a stabilised ±12 V supply; DC-coupled audio interface (or Expert Sleepers ES-9/ES-8 class) for CV generation and for recording the module's LFO if a CV out is present; otherwise probe the LFO LEDs optically or measure the sweep via the audio response.
- Capture set at 96 kHz / 24-bit, Depth = 0 (LFO off), Resonance = 0: swept sine 20 Hz–20 kHz at 1 V/oct CV = -3, -2, …, +5 V with PITCH centred; repeat with PITCH at min and max to map the knob. From each sweep, extract notch frequencies (NORMAL) and peak frequencies (ALTERNATIVE) → fit N and the pole offsets.
- Resonance characterisation: same sweeps at Resonance = 25/50/75/90/100 %; record self-oscillation onset, amplitude, waveform (FFT) and pitch vs. CV at 1 V steps.
- Nonlinearity: 1 kHz sine from -40 to +6 dBu at a static mid setting; THD+N vs level, and shift of the first notch frequency vs. level (this is the junction-resistance effect).
- LFO: record the swept response over 60 s at Rate min/mid/max with Depth = 100 %; derive the LFO waveform from the notch trajectory; repeat with SMOOTH on.
- Programme material (30 s drums, 30 s pad, 30 s noise) through each output and each COLOR setting for null tests.

6. Implementation requirements

- DSP core in Faust or C++ (JUCE/iPlug2). The all-pass chain is a trivially vectorisable structure; implement the 20 stages as a single state vector and update with the shared coefficient per sample (coefficient recomputed per sample or per 8-sample block from the smoothed control signal — measure the difference and pick the cheapest that passes Section 8).
- Feedback loop: resolve the implicit loop either with a one-sample delay (document the resulting pole shift and compensate) or with a zero-delay-feedback solve; the choice must be justified by the oscillation-pitch test in Section 8.
- 2× oversampling when Resonance > threshold or input level is high (nonlinearity engaged); none otherwise. Report latency = 0 to the host; oversampling filters must be zero-latency (IIR) or their latency reported.
- Formats: VST3, CLAP, AU; standalone WAV→WAV renderer for the measurement scripts.
- No Haible/Random*Source artwork, panel graphics, schematic excerpts or layout in the repo or UI.

7. Repository deliverables

- **README.md** with the credits in Section 9, a short description, build instructions, and the measured-fidelity table.
- **docs/**: this spec; block diagram; measurement protocol; results notebook with the fitted control law, notch map and oscillation data.
- **measure/**: Python scripts for notch extraction, control-law fit, THD+N, oscillation pitch tracking, null test; CI task that re-runs them and fails on regression.

- **src/**: DSP core with unit tests (single all-pass stage phase response, chain notch positions for ideal poles, feedback stability, LFO shape).
- **LICENSE**: MIT for code. Measurement data under CC0 or CC-BY (state which).

8. Verification protocol (acceptance criteria)

- Null test against the reference hardware at matched settings: record dry/wet through the same converter chain, align, subtract. Report residual in dB RMS and the residual spectrum. Target: better than -30 dB residual on a 2 s swept-sine and on a drum loop with LFO depth at zero (static settings), degrading gracefully with modulation.
- THD+N vs input level: 1 kHz sine from -40 dBFS to 0 dBFS in 5 dB steps, measured at the plugin output with the effect at a fixed static setting. Plot against the hardware curve.
- Aliasing: 10 kHz and 15 kHz sines at 0 dBFS, maximum modulation depth and rate; any alias component above -80 dBFS is a defect unless the hardware shows the same behaviour.
- CPU and latency budget: report per-instance CPU at 48 kHz / 64-sample buffers on one reference machine; algorithmic latency must be 0 samples (reported to the host as such); oversampling latency, if any, must be reported exactly.
- Every parameter has a documented unit and range in the spec table; the implementation exposes those units (ms, Hz, dB), not 0–127.
- All measurements are produced by scripts checked into the repo (Python, numpy/scipy) that take a WAV pair and emit the plots and numbers. No listening descriptions are accepted as evidence.

Device-specific additions

- 8.1 Notch map: with Depth = 0, the model's notch frequencies at five CV settings must match hardware within 2 % each; the number of notches below 20 kHz must match exactly (this fixes N).
- 8.2 Control law: fitted f_c(CV) must reproduce the hardware within 3 % across the full CV span, including any compression at the extremes caused by the 1/x stage.
- 8.3 Self-oscillation: onset resonance setting within 5 % of hardware; oscillation pitch vs. CV within ±25 cents over the range where the hardware itself tracks; oscillation waveform's first five harmonics within 3 dB.
- 8.4 Level dependence: shift of the first notch between -20 dBu and +6 dBu input must match hardware within 20 % of the measured shift.
- 8.5 Outputs: NORMAL vs ALTERNATIVE must be complementary as measured (peaks ↔ notches) and COLOR must invert exactly the measured path.
- 8.6 LFO: triangle shape (or measured shape) and rate limits within 5 %; SMOOTH corner frequency within 20 %.

9. Credits, trademarks and licensing (for README)

- **Jürgen Haible (1961–2011)**: designer of the 2007 redesign on which every modern Tau Pipe is based. Credit him by name and link his legacy page (jhaible.com/legacy/tau/jh_tau.html).
- **Dennis Colin and Ken McNeill / Tau Systems (Newton, NH)**: designers and maker of the original Tau "The Pipe" (developed 1975, released ≈1980).

- **Random*Source** (serge-modular.com, randomsource.net): states it holds all rights to Haible's electronic designs and is the only legitimate source for them; its documentation is marked "strictly for personal / educational use only" and asks not to be distributed or mirrored. The repo must not include any Random*Source or Haible schematic, BOM, PCB file or document; it may cite and link them. Before using "Haible" or "Tau Pipe" in the plugin's name, email Random*Source and ask; absent permission, use a descriptive name (e.g. "Pipe-20 Phaser") with a line: "an independent, non-commercial behavioural model inspired by the Tau 'The Pipe' phaser and Jürgen Haible's redesign; not affiliated with or endorsed by Random*Source or Tau Systems." Record their reply in the README.
- **ARP / Aries lineage**: acknowledge the ARP Quadra phaser relationship Haible cited, and cite US patent 3,969,682 (expired) as background.
- **Community documentation**: Bill and Will Hall (Dragonfly Alley) for the build documentation and US BOM; the electro-music.com DIY forum thread (2007 onward) where the design was developed in public; Scott Stites for the historical notes and PD-10 module information; keyboardmuseum.org for preserving the original "Pipe Phase Demo Record"; the PedalPCB build thread for the stompbox variant.
- **Prior DSP work**: cite any phaser-modelling or transistor-ladder papers you use (e.g. Huovilainen, Pirkle, Zavalishin's TPT/zero-delay-feedback method).
- Code: MIT. Note explicitly that the licence covers your code and data only, not the hardware designs it was measured against.

10. Open questions to resolve by measurement before coding

- Exact stage count on the reference unit (18 vs 20) and whether the Euro edition drops stages.
- Sign of the global feedback and whether the loop includes the dry signal.
- How ALTERNATIVE is derived (sign flip of wet vs. a mid-chain tap).
- The 1/x generator's effect on the sweep law at the extremes, and the knob taper of PITCH.
- LFO rate range, slow-mode range, and SMOOTH corner frequency.
- Level dependence magnitude with THAT300/BCM847 pairs (reference) vs. CA3046 builds (Haible's cheap option) — decide which the plugin's default models.

11. Sources

- Jürgen Haible, Tau Phaser legacy page: http://jhaible.com/legacy/tau/jh_tau.html
- Random*Source, Haible Tau "Pipe" Phaser (Euro) 2021 documentation: https://serge-modular.com/docs/Haible%20Tau%20Phaser%20eurorack%202021.pdf
- Random*Source, Haible Tau Pipe Phaser 2016 PCB page: https://serge-modular.com/haible/tau_phaser
- Perfect Circuit product page (18-pole description, I/O, power): https://www.perfectcircuit.com/random-source-haible-tau-phaser.html
- Matrixsynth, JHaible TAU Phaser 'The Pipe' 2026 edition: https://www.matrixsynth.com/2026/08/jhaible-tau-phaser-pipe-2026-by.html
- Matrixsynth, 20-pole phaser build notes (2024): https://www.matrixsynth.com/2024/07/20-pole-phaser-by-jurgen-haible-running.html
- Dragonfly Alley (Bill & Will Hall) construction notes: https://www.dragonflyalley.com/synth/constructionJHTauPipe.htm
- electro-music.com original 2007 project thread: https://electro-music.com/forum/post-121980.html
- DSL-man build log (MOTM kit, calibration notes): https://www.dsl-man.de/x/oQB4
- PedalPCB Tau Pipe V1.02 build thread: https://forum.pedalpcb.com/posts/377038/
- US Patent 3,969,682, Circuit for dynamic control of phase shift (1974): https://patents.google.com/patent/US3969682